

AWS IoT Analytics: AWS IoT Analytics enables you to collect, process, and analyse IoT data. It offers features for data ingestion, transformation, storage, and visualisation. You can use it to build real-time and batch analytics pipelines for IoT data.

AWS IoT Events: This service helps you detect and respond to IoT events and triggers actions based on defined rules. For example, you can use AWS IoT Events to trigger alerts or initiate automated workflows when certain conditions are met.

AWS IoT Greengrass: Greengrass extends AWS IoT to the edge, allowing you to run IoT applications and perform local processing on IoT devices. It enables devices to operate even when they are not connected to the internet, and provides features for local data caching and machine learning inference.

AWS IoT SiteWise: AWS IoT SiteWise is used for industrial IoT applications. It helps collect, structure, and analyse data from industrial equipment and processes, allowing you to monitor and optimise industrial operations.

AWS IoT Things Graph: This service lets you visually connect different devices and web services to build IoT applications using a drag-and-drop interface. It simplifies the integration of various IoT components.

AWS IoT Device Defender: Device Defender is a security service that continuously audits your IoT configurations and identifies potential security issues. It helps you monitor and protect your IoT environment from security threats.

AWS IoT FleetWise: This service is designed for fleet management, enabling you to manage and monitor large numbers of connected vehicles, such as cars, trucks, or drones.

Amazon FreeRTOS: Amazon FreeRTOS is an operating system for microcontrollers that makes it easy

to securely connect small, low-power devices to AWS IoT services.

AWS IoT 1-Click: This service simplifies the deployment of IoT devices with just one click, making it easier for customers to set up simple IoT solutions like dash buttons.

Using AWS IoT services for a smart farm

Let's now see how we can use the AWS services listed above for using IoT in a smart farm.

- Third-party drones send data through AWS Lambda for protocol conversion.
- Sensors or cameras running FreeRTOS send data to AWS IoT Greengrass, providing protection from intermittent connectivity.
- AWS IoT Greengrass streams enable ingestion from edge devices to Kinesis Data Streams.
- Real-time video via Amazon Kinesis Video Streams is used for streaming and replaying video content.
- Real-time insights are derived with Amazon Managed Service for Apache Flink and users are notified via Amazon Simple Notification Service.
- Analytics is enabled with OpenSearch, and Amazon Simple Storage Service is used for a data lake strategy.
- Owned data, like planting records or farm finances, is transferred

securely into your data lake with AWS Direct Connect.

- Data from a sensor ecosystem hosted on AWS with AWS PrivateLink is consumed securely.
 - Users are empowered with insights delivered via Amazon API Gateway or visualisations with Amazon QuickSight.
 - Machine learning (ML) models are built and deployed for edge inference with Amazon SageMaker. Amazon SageMaker Ground Truth is used to manage data labelling workflow.
 - Each time a new file is written into Amazon S3, AWS Glue Crawler crawls the data to infer the schema and make it available into the AWS Glue Data Catalog. Amazon Athena does on-demand querying.
 - A Lambda function is used to import the AWS IoT Device Defender reports into AWS Security Hub, in order to centralise incident response.
- IoT in agriculture is poised to bring about significant advancements and transformations in the way we cultivate, manage, and monitor crops. Implementing IoT in agriculture with AWS IoT services empowers farmers with the tools and insights needed to optimise their operations, reduce costs, improve crop quality, and make more informed decisions. **END** 

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By: Venkateswaran N.

The author is a lead architect, containerisation and modernisation, AWS Practice, IBM India.